Pediatric Cyanogenic Glycoside Exposures Reported to U.S. Poison Centers, 2005–2024

Originally accepted at the 46th EAPCCT Congress, 2026 · 20-year retrospective · N = 51,703

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BACKGROUND

Cyanogenic glycosides metabolize to hydrogen cyanide. They occur in cherry pits, apricot kernels, bitter almonds, cassava root, lima beans, and bamboo shoots. HCN release varies with whether the plant or seed is damaged or chewed.

RESEARCH QUESTION

How severe are unintentional pediatric cyanogenic-glycoside exposures, and what management strategy do these data support?

METHODS

- Retrospective NPDS analysis · 53 U.S. Poison Centers
- Period: January 2005 – December 2024
- Inclusion: < 6 years, single-substance ingestion
- N = 51,703 cyanogenic-plant ingestions

KEY FINDINGS

96.7%

MANAGED AT HOME

49,976 of 51,703 cases · No deaths reported

DETAILED RESULTS · N = 51,703

DISPOSITION · TABLE 1

96.7%
MANAGED AT HOME

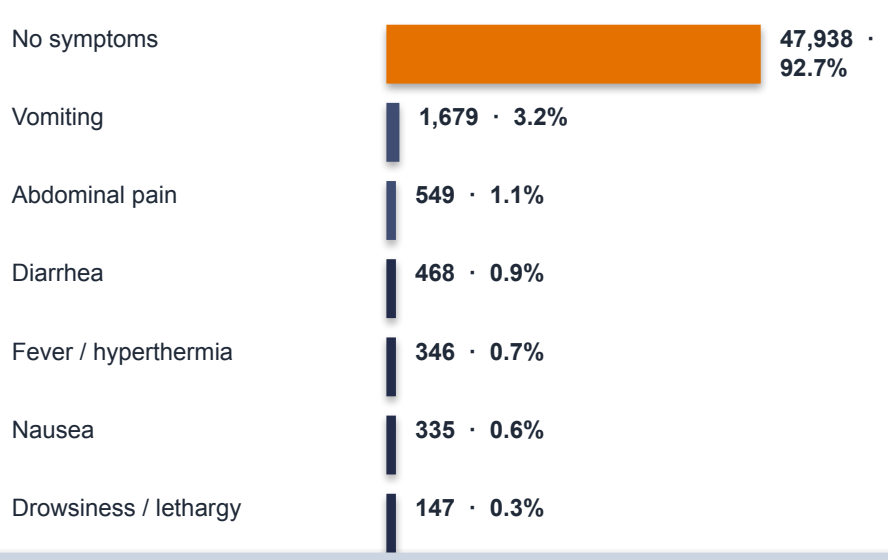
- **2.5%** TREATED & RELEASED — n = 1,294
- **0.07%** ADMITTED, NON-CRITICAL — n = 36
- **0.04%** ADMITTED, CRITICAL — n = 18

MEDICAL OUTCOMES · TABLE 2

47.7%
MINIMAL EFFECTS

- **26.1%** NON-TOXIC EXPOSURE — n = 13,505
- **21%** NO EFFECT — n = 10,844
- **2.35%** MINOR + MOD — n = 1,208
- **0.002%** MAJOR · 0 DEATHS — n = 1

CLINICAL EFFECTS · TABLE 3 · Top 7



Other common effects (each ≤0.05%):
CNS depression (92) · dizziness/vertigo (14) · seizures (6) · acidosis (5) · syncope (4) · coma (3) · hypotension (2) · cardiac arrest (1).

TREATMENTS · TABLE 4

Intervention	Number · %
Dilute / irrigate / wash	23,398 · 45.3%
Antiemetics	72 · 0.1%
IV fluids	48 · 0.1%
Oxygen	3 · 0.006%
Vasopressors	2 · 0.004%
Hydroxocobalamin	2 · 0.004%
Intubation	1 · 0.002%

KEY TAKEAWAYS

- 92.7% of cases had no clinical symptoms
- Hydroxocobalamin given in only 2 of 51,703 cases
- One major outcome; zero deaths reported in 20-year window
- Data support home management of asymptomatic ingestions

ASK ME ABOUT

- Triage criteria for asymptomatic pediatric exposures
- When to escalate to poison-center consult
- Why hydroxocobalamin was rarely indicated
- Plans for manuscript extension of this work